Tech Pay Shift — Global Compensation & Remote Work Dynamics (2020–2024)

Kaggle DS Salaries + World Bank GDP (PPP)

Interactive Visualization Project

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Motivation

How did the rapid expansion of remote work structurally reshape global tech compensation between 2020–2024?

Is salary growth homogeneous across roles and geographies, or does it exhibit structural divergence?

Does remote work enable cross-border purchasing power arbitrage by decoupling compensation from local GDP baselines?

Why must we move beyond averages and examine full distributional shifts to understand labor-market transformation?

The Data

Primary Dataset
Kaggle – Data Science Salaries (2020–2024)
https://www.kaggle.com/datasets/ruchi798/data-science-job-salaries

- 4,000+ global tech salary records
- Salary (USD standardized)
- Experience level
- Remote ratio (0%, 50%, 100%)
- Company location & employee residence

Macroeconomic Dataset
World Bank – GDP per Capita (PPP, Current International $)
https://data.worldbank.org/indicator/NY.GDP.PCAP.PP.CD

- Country-level GDP adjusted for purchasing power parity
- Enables cross-country economic normalization

Dataset Uniqueness

Integrated **micro-level salary data** with **macro-level economic indicators**

Constructed a derived metric: Salary-to-GDP (PPP) Ratio

Enables structural comparison of compensation relative to national economic baselines

Captures global remote labor-market decoupling

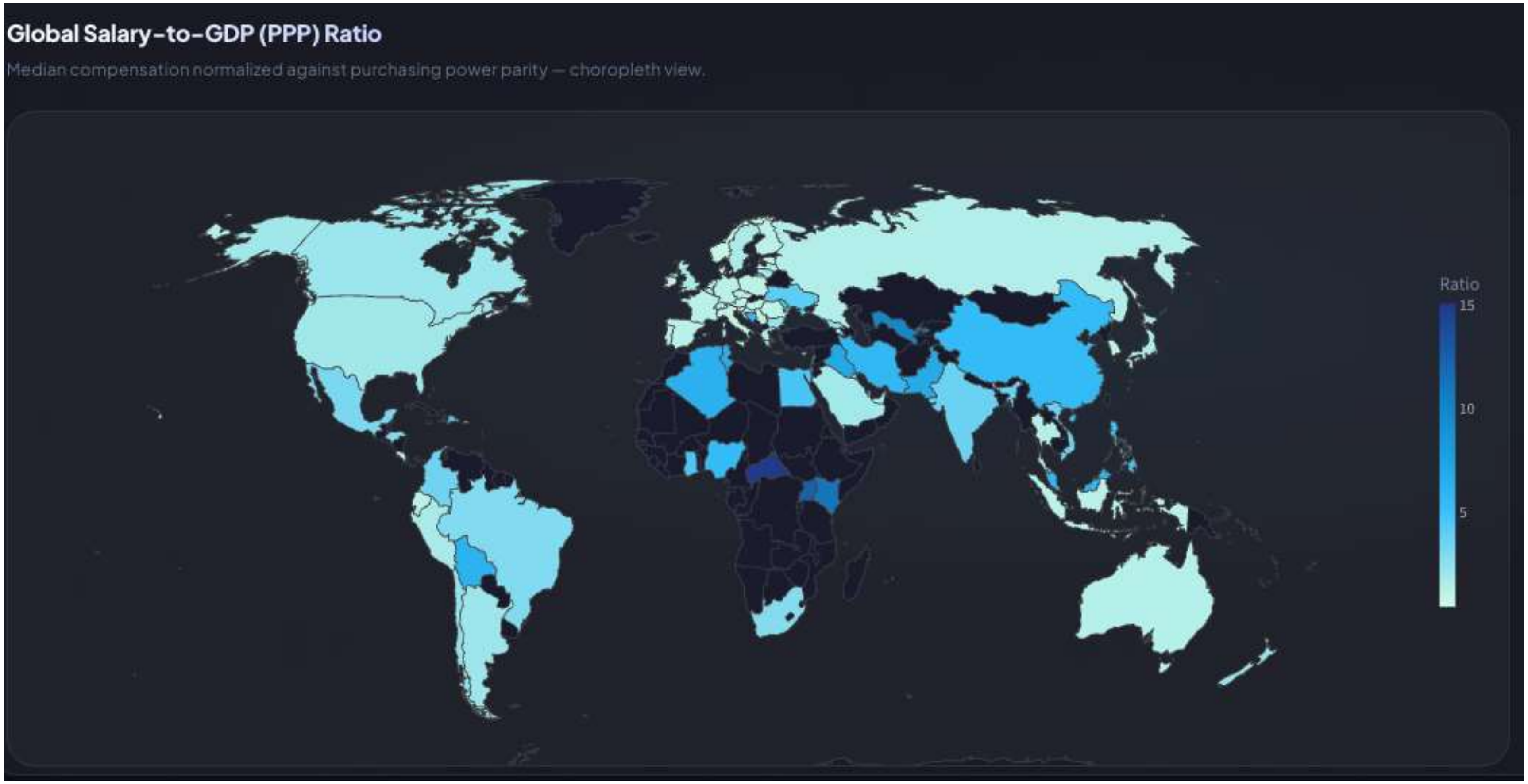
Multi-year window (2020–2024) allows post-pandemic temporal analysis

Process & Learning

- Shifted from relying on averages to analyzing full distributions, revealing inequality and upper-tail dynamics that summary statistics hide
- Applied PPP normalization to connect micro-level salary data with macroeconomic context, enabling meaningful global comparison
- Designed a structured, multi-tab interactive dashboard to guide users from distribution → geography → time divergence → inequality
- Learned that effective visualization is not just technical accuracy, it requires narrative clarity, intentional design, and audience awareness

Salary Distribution by Experience & Work Mode

Global Salary-to-GDP (PPP) Ratio



Time-Series Divergence: Remote vs Onsite

Distribution Evolution (2020–2024)

The Remote Compensation Story (2020–2024)

2020–2021
Remote work changed how companies paid tech workers.

2022–2023
High salaries grew fastest, especially at senior levels.

Emerging Markets
Remote jobs increased global earning power differences.

2024
Salary growth slowed and began to stabilize.

Audience & Impact

Who Is This For?

- **Policy Analysts**
  Studying global wage inequality and purchasing power dynamics
- **Tech Companies**
  Designing fair and competitive remote compensation models
- **Economists & Labor Researchers**
  Analyzing globalization and structural wage divergence
- **Students & Data Practitioners**
  Learning how to interpret distributional shifts beyond averages

Why It Matters

Remote work is not just a workplace trend, it is reshaping global compensation structures and economic balance.

Take Home Message

- Remote work structurally repriced global technical labor, reshaping compensation patterns across experience levels and geographies
- Salary growth was concentrated in upper distribution tiers, indicating divergence rather than uniform uplift
- Compensation became increasingly decoupled from national GDP baselines, enabling cross-border purchasing power effects
- Distribution-aware visualization reveals structural labor-market shifts more effectively than summary statistics alone